

T3 group factors: EFT_1_after_F

Active theory: SM + $S_1 + S_2 + C_{LLS_1S_2}$. The fermion F is absent, so y_1, y_2 and M_F are not running EFT couplings.

$$j = \frac{d-1}{2}, \quad C_2(d) = \frac{d^2-1}{4}, \quad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table, $\mathcal{S}$ denotes the RGE structure multiplying the displayed group factor in $16\pi^2\beta$.

Notation key

symbol	interaction / definition
g_1, g_2, g_3	SM gauge interactions
Y_u, Y_d, Y_e	SM Yukawa interactions
y_1	$LF S_1 + \text{h.c.}$
y_2	$LF S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2, m_2^2	$m_1^2 S_1^\dagger S_1, \ m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2} \lambda_1 (H^\dagger H)^2$
$\lambda_{S_i}^{(1)}$	$\frac{1}{2} \lambda_{S_i}^{(1)} (S_i^\dagger S_i)^2$
$\lambda_{HS_i}^{(1)}$	$(H^\dagger H)(S_i^\dagger S_i)$
$\lambda_{HS_i}^{(A)}$	$(H^\dagger T^A H)(S_i^\dagger T^A S_i)$
λ_5	$HH S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$(S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$(S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$(S_1^\dagger S_2)(S_2^\dagger S_1)$

Stage field content

run	S_1	S_2	F	integrated at this stage	active after matching
T3-A-m4	$(1, -2)$	$(3, -1)$	$(2, -\frac{3}{2})$	F	S1, S2
T3-A-m2	$(1, -1)$	$(3, 0)$	$(2, -\frac{1}{2})$	F	S1, S2
T3-A-p0	$(1, 0)$	$(3, 1)$	$(2, \frac{1}{2})$	F	S1, S2

T3-B-m3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	$(1, -1)$	F	S1, S2
T3-B-m1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	$(1, 0)$	F	S1, S2
T3-B-p1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	$(1, 1)$	F	S1, S2
T3-C-m3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	$(3, -1)$	F	S1, S2
T3-C-m1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	$(3, 0)$	F	S1, S2
T3-C-p1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	$(3, 1)$	F	S1, S2
T3-D-m2	$(3, -1)$	$(1, 0)$	$(2, -\frac{1}{2})$	F	S1, S2
T3-D-p0	$(3, 0)$	$(1, 1)$	$(2, \frac{1}{2})$	F	S1, S2
T3-D-p2	$(3, 1)$	$(1, 2)$	$(2, \frac{3}{2})$	F	S1, S2
T3-E-m4	$(3, -2)$	$(3, -1)$	$(2, -\frac{3}{2})$	F	S1, S2
T3-E-m2	$(3, -1)$	$(3, 0)$	$(2, -\frac{1}{2})$	F	S1, S2
T3-E-p0	$(3, 0)$	$(3, 1)$	$(2, \frac{1}{2})$	F	S1, S2
T3-E-p2	$(3, 1)$	$(3, 2)$	$(2, \frac{3}{2})$	F	S1, S2

Field representations

run	S_1	S_2	α
T3 - A - m4	$(1, -2)$	$(3, -1)$	-4
T3 - A - m2	$(1, -1)$	$(3, 0)$	-2
T3 - A - p0	$(1, 0)$	$(3, 1)$	0
T3 - B - m3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	-3
T3 - B - m1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	-1
T3 - B - p1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	1
T3 - C - m3	$(2, -\frac{3}{2})$	$(2, -\frac{1}{2})$	-3
T3 - C - m1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	-1
T3 - C - p1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	1
T3 - D - m2	$(3, -1)$	$(1, 0)$	-2
T3 - D - p0	$(3, 0)$	$(1, 1)$	0
T3 - D - p2	$(3, 1)$	$(1, 2)$	2
T3 - E - m4	$(3, -2)$	$(3, -1)$	-4
T3 - E - m2	$(3, -1)$	$(3, 0)$	-2
T3 - E - p0	$(3, 0)$	$(3, 1)$	0
T3 - E - p2	$(3, 1)$	$(3, 2)$	2

Gauge and Yukawa beta functions in this EFT

$$\beta_{g_1}^{(1)}$$

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
g_1^3	$\frac{55}{6}$	$\frac{43}{6}$	$\frac{47}{6}$	$\frac{17}{2}$	$\frac{43}{6}$

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
g_1^3	$\frac{17}{2}$	$\frac{17}{2}$	$\frac{43}{6}$	$\frac{17}{2}$	$\frac{47}{6}$

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
g_1^3	$\frac{43}{6}$	$\frac{55}{6}$	$\frac{71}{6}$	$\frac{47}{6}$	$\frac{47}{6}$

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
g_1^3	$\frac{71}{6}$

$$\beta_{g_2}^{(1)}$$

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
g_2^3	$-\frac{5}{2}$	$-\frac{5}{2}$	$-\frac{5}{2}$	$-\frac{17}{6}$	$-\frac{17}{6}$

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
g_2^3	$-\frac{17}{6}$	$-\frac{17}{6}$	$-\frac{17}{6}$	$-\frac{17}{6}$	$-\frac{5}{2}$

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
g_2^3	$-\frac{5}{2}$	$-\frac{5}{2}$	$-\frac{11}{6}$	$-\frac{11}{6}$	$-\frac{11}{6}$

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
g_2^3	$-\frac{11}{6}$

$$\beta_{g_3}^{(1)}$$

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
g_3^3	-7	-7	-7	-7	-7

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
g_3^3	-7	-7	-7	-7	-7

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
g_3^3	-7	-7	-7	-7	-7

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
g_3^3	-7

$$\beta_{Y_u}^{(1)}$$

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1

$$\beta_{Y_d}^{(1)}$$

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8	-8	-8	-8	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8	-8	-8	-8	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8	-8	-8	-8	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1

$$\beta_{Y_e}^{(1)}$$

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$

$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1

$\beta_{m_1^2}$

Run block 1: T3-A-m4-T3-B-m1

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$\lambda_{S_1}^{(1)} m_1^2$	4	4	4	6	6
$g_2^2 m_1^2$	0	0	0	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2 m_1^2$	-24	-6	0	$-\frac{27}{2}$	$-\frac{3}{2}$
$\lambda_{12}^{(1)} m_2^2$	6	6	6	4	4
$\lambda_{S_1}^{(A)} m_1^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} m_2^2$	0	0	0	0	0

Run block 2: T3-B-p1-T3-D-m2

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$\lambda_{S_1}^{(1)} m_1^2$	6	6	6	6	8
$g_2^2 m_1^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	-12
$g_1^2 m_1^2$	$-\frac{3}{2}$	$-\frac{27}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	-6
$\lambda_{12}^{(1)} m_2^2$	4	4	4	4	2
$\lambda_{S_1}^{(A)} m_1^2$	0	0	0	0	4
$\lambda_{12}^{(\times)} m_2^2$	0	0	0	0	0

Run block 3: T3-D-p0-T3-E-p0

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$\lambda_{S_1}^{(1)} m_1^2$	8	8	8	8	8
$g_2^2 m_1^2$	-12	-12	-12	-12	-12

$g_1^2 m_1^2$	0	−6	−24	−6	0
$\lambda_{12}^{(1)} m_2^2$	2	2	6	6	6
$\lambda_{S_1}^{(A)} m_1^2$	4	4	4	4	4
$\lambda_{12}^{(\times)} m_2^2$	0	0	4	4	4

Run block 4: $T3\text{-}E\text{-}p2\text{-}T3\text{-}E\text{-}p2$

$\mathcal{S}$	T3 − E − p2 ($\alpha = 2$)
$\lambda_{S_1}^{(1)} m_1^2$	8
$g_2^2 m_1^2$	−12
$g_1^2 m_1^2$	−6
$\lambda_{12}^{(1)} m_2^2$	6
$\lambda_{S_1}^{(A)} m_1^2$	4
$\lambda_{12}^{(\times)} m_2^2$	4

$\beta_{m_2^2}$

Run block 1: $T3\text{-}A\text{-}m4\text{-}T3\text{-}B\text{-}m1$

$\mathcal{S}$	T3 − A − m4 ($\alpha = -4$)	T3 − A − m2 ($\alpha = -2$)	T3 − A − p0 ($\alpha = 0$)	T3 − B − m3 ($\alpha = -3$)	T3 − B − m1 ($\alpha = -1$)
$\lambda_{S_2}^{(1)} m_2^2$	8	8	8	6	6
$g_2^2 m_2^2$	−12	−12	−12	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2 m_2^2$	−6	0	−6	$-\frac{3}{2}$	$-\frac{3}{2}$
$\lambda_{12}^{(1)} m_1^2$	2	2	2	4	4
$\lambda_{S_2}^{(A)} m_2^2$	4	4	4	0	0
$\lambda_{12}^{(\times)} m_1^2$	0	0	0	0	0

Run block 2: $T3\text{-}B\text{-}p1\text{-}T3\text{-}D\text{-}m2$

$\mathcal{S}$	T3 − B − p1 ($\alpha = 1$)	T3 − C − m3 ($\alpha = -3$)	T3 − C − m1 ($\alpha = -1$)	T3 − C − p1 ($\alpha = 1$)	T3 − D − m2 ($\alpha = -2$)
$\lambda_{S_2}^{(1)} m_2^2$	6	6	6	6	4
$g_2^2 m_2^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	0
$g_1^2 m_2^2$	$-\frac{27}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{27}{2}$	0
$\lambda_{12}^{(1)} m_1^2$	4	4	4	4	6
$\lambda_{S_2}^{(A)} m_2^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} m_1^2$	0	0	0	0	0

Run block 3: T3-D-p0-T3-E-p0

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$\lambda_{S_2}^{(1)} m_2^2$	4	4	8	8	8
$g_2^2 m_2^2$	0	0	-12	-12	-12
$g_1^2 m_2^2$	-6	-24	-6	0	-6
$\lambda_{12}^{(1)} m_1^2$	6	6	6	6	6
$\lambda_{S_2}^{(A)} m_2^2$	0	0	4	4	4
$\lambda_{12}^{(\times)} m_1^2$	0	0	4	4	4

Run block 4: T3-E-p2-T3-E-p2

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$\lambda_{S_2}^{(1)} m_2^2$	8
$g_2^2 m_2^2$	-12
$g_1^2 m_2^2$	-24
$\lambda_{12}^{(1)} m_1^2$	6
$\lambda_{S_2}^{(A)} m_2^2$	4
$\lambda_{12}^{(\times)} m_1^2$	4

$\beta_{\lambda_{HS_1}^{(1)}}$

Run block 1: T3-A-m4-T3-B-m1

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$ \lambda_5 ^2$	12	12	12	6	6
$g_2^2 \lambda_{HS_1}^{(1)}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	-9	-9
$g_1^2 \lambda_{HS_1}^{(1)}$	$-\frac{51}{2}$	$-\frac{15}{2}$	$-\frac{3}{2}$	-15	-3
$\lambda_1 \lambda_{HS_1}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$	4	4	4	6	6
$\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$	6	6	6	4	4
g_2^4	0	0	0	$\frac{9}{4}$	$\frac{9}{4}$
g_1^4	12	3	0	$\frac{27}{4}$	$\frac{3}{4}$
$\lambda_{HS_1}^{(1)} T$	2	2	2	2	2
$\left(\lambda_{HS_1}^{(A)}\right)^2$	0	0	0	$\frac{3}{4}$	$\frac{3}{4}$

$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\lambda_{HS_2}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 2: T3-B-p1-T3-D-m2

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$ \lambda_5 ^2$	6	6	6	6	4
$g_2^2 \lambda_{HS_1}^{(1)}$	-9	-9	-9	-9	$-\frac{33}{2}$
$g_1^2 \lambda_{HS_1}^{(1)}$	-3	-15	-3	-3	$-\frac{15}{2}$
$\lambda_1 \lambda_{HS_1}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$	6	6	6	6	8
$\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$	4	4	4	4	2
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	6
g_1^4	$\frac{3}{4}$	$\frac{27}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	3
$\lambda_{HS_1}^{(1)} T$	2	2	2	2	2
$\left(\lambda_{HS_1}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	2
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(A)}$	0	0	0	0	4
$\lambda_{HS_2}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 3: T3-D-p0-T3-E-p0

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$ \lambda_5 ^2$	4	4	4	4	4
$g_2^2 \lambda_{HS_1}^{(1)}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2 \lambda_{HS_1}^{(1)}$	$-\frac{3}{2}$	$-\frac{15}{2}$	$-\frac{51}{2}$	$-\frac{15}{2}$	$-\frac{3}{2}$
$\lambda_1 \lambda_{HS_1}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$	8	8	8	8	8
$\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$	2	2	6	6	6
g_2^4	6	6	6	6	6
g_1^4	0	3	12	3	0
$\lambda_{HS_1}^{(1)} T$	2	2	2	2	2
$\left(\lambda_{HS_1}^{(A)}\right)^2$	2	2	2	2	2
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(A)}$	4	4	4	4	4

$\lambda_{HS_2}^{(1)} \lambda_{12}^{(\times)}$	0	0	4	4	4
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Run block 4: T3-E-p2-T3-E-p2

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$ \lambda_5 ^2$	4
$g_2^2 \lambda_{HS_1}^{(1)}$	$-\frac{33}{2}$
$g_1^2 \lambda_{HS_1}^{(1)}$	$-\frac{15}{2}$
$\lambda_1 \lambda_{HS_1}^{(1)}$	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$	8
$\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$	6
g_2^4	6
g_1^4	3
$\lambda_{HS_1}^{(1)} T$	2
$\left(\lambda_{HS_1}^{(A)}\right)^2$	2
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(A)}$	4
$\lambda_{HS_2}^{(1)} \lambda_{12}^{(\times)}$	4

$\beta_{\lambda_{HS_2}^{(1)}}$

Run block 1: T3-A-m4-T3-B-m1

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$ \lambda_5 ^2$	4	4	4	6	6
$g_2^2 \lambda_{HS_2}^{(1)}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	−9	−9
$g_1^2 \lambda_{HS_2}^{(1)}$	$-\frac{15}{2}$	$-\frac{3}{2}$	$-\frac{15}{2}$	−3	−3
$\lambda_1 \lambda_{HS_2}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$	8	8	8	6	6
$\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$	2	2	2	4	4
g_2^4	6	6	6	$\frac{9}{4}$	$\frac{9}{4}$
g_1^4	3	0	3	$\frac{3}{4}$	$\frac{3}{4}$
$\left(\lambda_{HS_2}^{(A)}\right)^2$	2	2	2	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(A)}$	4	4	4	0	0

$\lambda_{HS_2}^{(1)} T$	2	2	2	2	2
$\lambda_{HS_1}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 2: T3-B-p1-T3-D-m2

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$ \lambda_5 ^2$	6	6	6	6	12
$g_2^2 \lambda_{HS_2}^{(1)}$	-9	-9	-9	-9	$-\frac{9}{2}$
$g_1^2 \lambda_{HS_2}^{(1)}$	-15	-3	-3	-15	$-\frac{3}{2}$
$\lambda_1 \lambda_{HS_2}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$	6	6	6	6	4
$\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$	4	4	4	4	6
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	0
g_1^4	$\frac{27}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{27}{4}$	0
$\left(\lambda_{HS_2}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	0
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(A)}$	0	0	0	0	0
$\lambda_{HS_2}^{(1)} T$	2	2	2	2	2
$\lambda_{HS_1}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 3: T3-D-p0-T3-E-p0

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$ \lambda_5 ^2$	12	12	4	4	4
$g_2^2 \lambda_{HS_2}^{(1)}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2 \lambda_{HS_2}^{(1)}$	$-\frac{15}{2}$	$-\frac{51}{2}$	$-\frac{15}{2}$	$-\frac{3}{2}$	$-\frac{15}{2}$
$\lambda_1 \lambda_{HS_2}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$	4	4	8	8	8
$\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$	6	6	6	6	6
g_2^4	0	0	6	6	6
g_1^4	3	12	3	0	3
$\left(\lambda_{HS_2}^{(A)}\right)^2$	0	0	2	2	2
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(A)}$	0	0	4	4	4
$\lambda_{HS_2}^{(1)} T$	2	2	2	2	2

$\lambda_{HS_1}^{(1)} \lambda_{12}^{(\times)}$	0	0	4	4	4
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Run block 4: T3-E-p2-T3-E-p2

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$ \lambda_5 ^2$	4
$g_2^2 \lambda_{HS_2}^{(1)}$	$-\frac{33}{2}$
$g_1^2 \lambda_{HS_2}^{(1)}$	$-\frac{51}{2}$
$\lambda_1 \lambda_{HS_2}^{(1)}$	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$	8
$\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$	6
g_2^4	6
g_1^4	12
$\left(\lambda_{HS_2}^{(A)}\right)^2$	2
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(A)}$	4
$\lambda_{HS_2}^{(1)} T$	2
$\lambda_{HS_1}^{(1)} \lambda_{12}^{(\times)}$	4

$$\beta_{\lambda_{12}^{(1)}}$$

Run block 1: T3-A-m4-T3-B-m1

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$ \lambda_5 ^2$	4	4	4	3	3
$g_2^2 \lambda_{12}^{(1)}$	-12	-12	-12	-9	-9
$g_1^2 \lambda_{12}^{(1)}$	-30	-6	-6	-15	-3
$\left(\lambda_{12}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$	4	4	4	6	6
$\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$	8	8	8	6	6
$\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$	4	4	4	4	4
g_2^4	0	0	0	$\frac{9}{4}$	$\frac{9}{4}$
g_1^4	48	0	0	$\frac{27}{4}$	$\frac{3}{4}$
$\lambda_{12}^{(1)} \lambda_{S_2}^{(A)}$	4	4	4	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	0	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_{12}^{(1)} \lambda_{S_1}^{(A)}$	0	0	0	0	0

$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	0	0	0

Run block 2: T3-B-p1-T3-D-m2

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$ \lambda_5 ^2$	3	3	3	3	4
$g_2^2 \lambda_{12}^{(1)}$	−9	−9	−9	−9	−12
$g_1^2 \lambda_{12}^{(1)}$	−15	−15	−3	−15	−6
$\left(\lambda_{12}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$	6	6	6	6	8
$\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$	6	6	6	6	4
$\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$	4	4	4	4	4
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	0
g_1^4	$\frac{27}{4}$	$\frac{27}{4}$	$\frac{3}{4}$	$\frac{27}{4}$	0
$\lambda_{12}^{(1)} \lambda_{S_2}^{(A)}$	0	0	0	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	0
$\lambda_{12}^{(1)} \lambda_{S_1}^{(A)}$	0	0	0	0	4
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	0	0	0

Run block 3: T3-D-p0-T3-E-p0

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$ \lambda_5 ^2$	4	4	$\frac{4}{3}$	$\frac{4}{3}$	$\frac{4}{3}$
$g_2^2 \lambda_{12}^{(1)}$	−12	−12	−24	−24	−24
$g_1^2 \lambda_{12}^{(1)}$	−6	−30	−30	−6	−6
$\left(\lambda_{12}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$	8	8	8	8	8
$\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$	4	4	8	8	8
$\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$	4	4	4	4	4
g_2^4	0	0	12	12	12
g_1^4	0	48	48	0	0

$\lambda_{12}^{(1)} \lambda_{S_2}^{(A)}$	0	0	4	4	4
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	4	4	4
$\lambda_{12}^{(1)} \lambda_{S_1}^{(A)}$	4	4	4	4	4
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	4	4	4
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	4	4	4
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	4	4	4

Run block 4: T3-E-p2-T3-E-p2

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$ \lambda_5 ^2$	$\frac{4}{3}$
$g_2^2 \lambda_{12}^{(1)}$	−24
$g_1^2 \lambda_{12}^{(1)}$	−30
$\left(\lambda_{12}^{(1)}\right)^2$	4
$\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$	8
$\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$	8
$\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$	4
g_2^4	12
g_1^4	48
$\lambda_{12}^{(1)} \lambda_{S_2}^{(A)}$	4
$\left(\lambda_{12}^{(A)}\right)^2$	4
$\lambda_{12}^{(1)} \lambda_{S_1}^{(A)}$	4
$\left(\lambda_{12}^{(\times)}\right)^2$	4
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	4
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	4

β_{λ_5}

Run block 1: T3-A-m4-T3-B-m1

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$\lambda_5 \lambda_1$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{HS_2}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{12}^{(1)}$	2	2	2	2	2

$g_2^2 \lambda_5$	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{21}{2}$	-9	-9
$g_1^2 \lambda_5$	$-\frac{33}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	-9	-3
$\lambda_5 \lambda_{HS_2}^{(A)}$	4	4	4	2	2
$\lambda_5 T$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(A)}$	0	0	0	-2	-2
$\lambda_5 \lambda_{12}^{(A)}$	0	0	0	$-\frac{1}{2}$	$-\frac{1}{2}$
$\lambda_5 \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 2: T3-B-p1-T3-D-m2

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$\lambda_5 \lambda_1$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{HS_2}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{12}^{(1)}$	2	2	2	2	2
$g_2^2 \lambda_5$	-9	-9	-9	-9	$-\frac{21}{2}$
$g_1^2 \lambda_5$	-9	-9	-3	-9	$-\frac{9}{2}$
$\lambda_5 \lambda_{HS_2}^{(A)}$	2	2	2	2	0
$\lambda_5 T$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(A)}$	-2	-2	-2	-2	-4
$\lambda_5 \lambda_{12}^{(A)}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	0
$\lambda_5 \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 3: T3-D-p0-T3-E-p0

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$\lambda_5 \lambda_1$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{HS_2}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{12}^{(1)}$	2	2	2	2	2
$g_2^2 \lambda_5$	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2 \lambda_5$	$-\frac{9}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$\lambda_5 \lambda_{HS_2}^{(A)}$	0	0	2	2	2
$\lambda_5 T$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(A)}$	-4	-4	-2	-2	-2
$\lambda_5 \lambda_{12}^{(A)}$	0	0	2	2	2
$\lambda_5 \lambda_{12}^{(\times)}$	0	0	-2	-2	-2

Run block 4: $T3\text{-}E\text{-}p2\text{-}T3\text{-}E\text{-}p2$

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$\lambda_5 \lambda_1$	2
$\lambda_5 \lambda_{HS_1}^{(1)}$	4
$\lambda_5 \lambda_{HS_2}^{(1)}$	4
$\lambda_5 \lambda_{12}^{(1)}$	2
$g_2^2 \lambda_5$	$-\frac{33}{2}$
$g_1^2 \lambda_5$	$-\frac{33}{2}$
$\lambda_5 \lambda_{HS_2}^{(A)}$	2
$\lambda_5 T$	2
$\lambda_5 \lambda_{HS_1}^{(A)}$	−2
$\lambda_5 \lambda_{12}^{(A)}$	2
$\lambda_5 \lambda_{12}^{(\times)}$	−2

Additional scalar-coupling beta functions

$\beta_{\lambda_{HS_1}^{(A)}}$

Run block 1: $T3\text{-}A\text{-}m4\text{-}T3\text{-}B\text{-}m1$

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$g_1^2 g_2^2$	0	0	0	−18	−6

Run block 2: $T3\text{-}B\text{-}p1\text{-}T3\text{-}D\text{-}m2$

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$g_1^2 g_2^2$	6	−18	−6	6	−12

Run block 3: $T3\text{-}D\text{-}p0\text{-}T3\text{-}E\text{-}p0$

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$g_1^2 g_2^2$	0	12	−24	−12	0

Run block 4: $T3\text{-}E\text{-}p2\text{-}T3\text{-}E\text{-}p2$

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$g_1^2 g_2^2$	12

$$\beta_{\lambda_{HS_2}^{(A)}}$$

Run block 1: T3-A-m4-T3-B-m1

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$g_1^2 g_2^2$	-12	0	12	-6	6

Run block 2: T3-B-p1-T3-D-m2

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$g_1^2 g_2^2$	18	-6	6	18	0

Run block 3: T3-D-p0-T3-E-p0

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$g_1^2 g_2^2$	0	0	-12	0	12

Run block 4: T3-E-p2-T3-E-p2

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$g_1^2 g_2^2$	24

$$\beta_{\lambda_{12}^{(A)}}$$

Run block 1: T3-A-m4-T3-B-m1

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$g_1^2 g_2^2$	0	0	0	18	-6

Run block 2: T3-B-p1-T3-D-m2

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$g_1^2 g_2^2$	18	18	-6	18	0

Run block 3: T3-D-p0-T3-E-p0

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$g_1^2 g_2^2$	0	0	48	0	0

Run block 4: T3-E-p2-T3-E-p2

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$g_1^2 g_2^2$	48

$\beta_{\lambda_{12}^{(\times)}}$

Run block 1: T3-A-m4-T3-B-m1

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
g_2^4	0	0	0	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	0	0	0

Run block 2: T3-B-p1-T3-D-m2

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
g_2^4	0	0	0	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	0	0	0

$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	0	0	0
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Run block 3: T3-D-p0-T3-E-p0

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
g_2^4	0	0	6	6	6
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	2	2	2
$\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	0	0	8	8	8
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	10	10	10
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	−2	−2	−2
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	−2	−2	−2

Run block 4: T3-E-p2-T3-E-p2

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
g_2^4	6
$\left(\lambda_{12}^{(A)}\right)^2$	2
$\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	8
$\left(\lambda_{12}^{(\times)}\right)^2$	10
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	−2
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	−2

Full Weinberg-coefficient RGE in EFT1

Writing the canonical mixed LLSS coefficient as C_{12} , the coupled one-loop Weinberg equation used in the F -first EFT is

$$16\pi^2 \frac{dC_5}{d\ln\mu} = (2\lambda_1 - 3g_2^2 + 2T)C_5 \quad (1)$$

$$- \frac{3}{2} [(Y_e^\dagger Y_e)^T C_5 + C_5 (Y_e^\dagger Y_e)] \quad (2)$$

$$+ R_W \lambda_5 C_{12}. \quad (3)$$

Note: $\frac{7}{2}$ thing isnt there if we integrate out our fermion F so its not here

Direct $C_{12} \rightarrow C_5$ recoupling factor

For each representation the source term is

$$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{12}.$$

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
R_W	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
R_W	$\frac{4\sqrt{3}}{3}$	$-\frac{4}{3}$	$-\frac{4}{3}$	$-\frac{4}{3}$	$\frac{4\sqrt{3}}{3}$

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
R_W	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$	$-\frac{4\sqrt{6}}{3}$	$-\frac{4\sqrt{6}}{3}$	$-\frac{4\sqrt{6}}{3}$

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
R_W	$-\frac{4\sqrt{6}}{3}$

.

T3 – A – m4 ($\alpha = -4$)

1 independent complex matched structure displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$\frac{\overline{y_1 y_2}}{M_F}$	$\mathcal{I}_{LLS_1 S_2}$

T3 – A – m2 ($\alpha = -2$)

1 independent complex matched structure displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$\frac{\overline{y_1 y_2}}{M_F}$	$\mathcal{I}_{LLS_1 S_2}$

T3 – A – p0 ($\alpha = 0$)

1 independent complex matched structure displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$\frac{\overline{y_1 y_2}}{M_F}$	$\mathcal{I}_{LLS_1 S_2}$

T3 – B – m3 ($\alpha = -3$)

1 independent complex matched structure displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$\frac{\overline{y_1 y_2}}{M_F}$	$\mathcal{C}_{LS_1 F} \mathcal{C}_{LS_2 F}$

T3 – B – m1 ($\alpha = -1$)

3 independent complex matched structures displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j) S_1^\dagger S_1^\dagger + \text{h.c.}$	$\frac{\overline{y_1}^2}{2M_F}$	$\mathcal{C}_{LS_1 F}$
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$\frac{\overline{y_1 y_2}}{M_F}$	$\mathcal{C}_{LS_1 F} \mathcal{C}_{LS_2 F}$
$(L_i^T C L_j) S_2 S_2 + \text{h.c.}$	$\frac{\overline{y_2}^2}{2M_F}$	$\mathcal{C}_{LS_2 F}$

T3 – B – p1 ($\alpha = 1$)

1 independent complex matched structure displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$\frac{\overline{y_1 y_2}}{M_F}$	$\mathcal{C}_{LS_1 F} \mathcal{C}_{LS_2 F}$

T3 – C – m3 ($\alpha = -3$)

2 independent complex matched structures displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$-\frac{\sqrt{3}\overline{y_1 y_2}}{2M_F}$	ϵ
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$-\frac{\overline{y_1 y_2}}{2M_F}$	$\mathcal{I}_{LLS_1 S_2}$

T3 – C – m1 ($\alpha = -1$)

4 independent complex matched structures displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j) S_1^\dagger S_1^\dagger + \text{h.c.}$	$\frac{\overline{y_1}^2}{2M_F}$	cg2
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$-\frac{\sqrt{3}\overline{y_1}y_2}{2M_F}$	ϵ
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$-\frac{\overline{y_1}y_2}{2M_F}$	$\mathcal{I}_{LLS_1S_2}$
$(L_i^T C L_j) S_2 S_2 + \text{h.c.}$	$\frac{\overline{y_2}^2}{2M_F}$	cg1

T3 – C – p1 ($\alpha = 1$)

2 independent complex matched structures displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$-\frac{\sqrt{3}\overline{y_1}y_2}{2M_F}$	ϵ
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$-\frac{\overline{y_1}y_2}{2M_F}$	$\mathcal{I}_{LLS_1S_2}$

T3 – D – m2 ($\alpha = -2$)

1 independent complex matched structure displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$\frac{\overline{y_1}y_2}{M_F}$	$\mathcal{I}_{LLS_1S_2}$

T3 – D – p0 ($\alpha = 0$)

1 independent complex matched structure displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$\frac{\overline{y_1}y_2}{M_F}$	$\mathcal{I}_{LLS_1S_2}$

T3 – D – p2 ($\alpha = 2$)

1 independent complex matched structure displayed.

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complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$\frac{\overline{y_1 y_2}}{M_F}$	$\mathcal{I}_{LLS_1 S_2}$

T3 – E – m4 ($\alpha = -4$)

2 independent complex matched structures displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$\frac{\sqrt{3} \overline{y_1 y_2}}{3 M_F}$	ϵ
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$\frac{\sqrt{6} \overline{y_1 y_2}}{3 M_F}$	$\mathcal{I}_{LLS_1 S_2}$

T3 – E – m2 ($\alpha = -2$)

2 independent complex matched structures displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$\frac{\sqrt{3} \overline{y_1 y_2}}{3 M_F}$	ϵ
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$\frac{\sqrt{6} \overline{y_1 y_2}}{3 M_F}$	$\mathcal{I}_{LLS_1 S_2}$

T3 – E – p0 ($\alpha = 0$)

2 independent complex matched structures displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$\frac{\sqrt{3} \overline{y_1 y_2}}{3 M_F}$	ϵ
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$\frac{\sqrt{6} \overline{y_1 y_2}}{3 M_F}$	$\mathcal{I}_{LLS_1 S_2}$

T3 – E – p2 ($\alpha = 2$)

2 independent complex matched structures displayed.

complex matched operator	matched prefactor	$SU(2)_L$ invariant
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$\frac{\sqrt{3} \overline{y_1 y_2}}{3 M_F}$	ϵ
$(L_i^T C L_j) S_1^\dagger S_2 + \text{h.c.}$	$\frac{\sqrt{6} \overline{y_1 y_2}}{3 M_F}$	$\mathcal{I}_{LLS_1 S_2}$

Next threshold

Further matching depends on the next field or fields selected by the threshold plan. Once all T3 fields are removed, their representation dependence is carried by the matched C_5 boundary condition and the subsequent SMEFT running is universal.